Human Review: convex-block homogeneous misreading

Packet identifier: `1711.01340_ell2_difference_basis_not_submultiplicative`
Original automatic proof: `solution_packet.pdf`
Checked date: 14 June 2026
Status: Manually checked.
Verdict: Rejected as a solution to the intended problem. This is a miss.

Human Comments

The proposed solution in `solution_packet.pdf` should not be accepted as a counterexample to Problem 7 of Motakis–Puglisi–Tolias. The finite calculation with the difference sequence in ℓ_2 is correct, but the example does not satisfy the hypothesis of the problem.

The packet claims that the usual unit vector basis of ℓ_2 is convex block homogeneous because every normalized block sequence is orthonormal. This uses the wrong notion. In the context cited by the source paper, Argyros–Motakis–Sari introduce a convex block homogeneous sequence as one which is equivalent to its convex block sequences. The convex blocks are not renormalized.

For the unit vector basis of ℓ_2, take successive finite sets B_k with $|B_k| \to \infty$, and define

$$y_k = \frac{1}{|B_k|} \sum_{i \in B_k} e_i.$$

Then (y_k) is a convex block sequence of the unit vector basis, but

$$\|y_k\|_2 = |B_k|^{-1/2} \longrightarrow 0.$$

Therefore (y_k) is not even seminormalized and cannot be equivalent to the normalized unit vector basis of ℓ_2. Thus the proposed example fails the defining hypothesis "convex block homogeneous basis".

This also fits the surrounding context of the source paper. Remark 8.10 mentions the known convex block homogeneous examples arising from $J(X)$ and $J^*(X)$, and then says that a positive answer to Problem 7 would imply a statement about difference bases of conditional spreading sequences. The question is not aimed at the Hilbert unit vector basis under a normalized-block interpretation.

There is a secondary warning as well: in ℓ_2, the sequence

$$d_1 = e_1, \qquad d_{i+1} = e_{i+1} - e_i$$

is not a Schauder basis of ℓ_2. The inverse coordinate map involves partial summation and is unbounded. This reinforces that the packet is testing a coefficient norm outside the intended setting.

Short Version

This is a miss. The packet treats "convex block homogeneous" as equivalence to normalized block sequences, but the source context uses equivalence to convex block sequences themselves. The ℓ_2 unit vector basis fails that hypothesis by taking long equal-average convex blocks.

Review Outcome

Reject this candidate as a proposed counterexample to Problem 7. It may be archived as a warning that automatic extraction must preserve the precise definition of "convex block homogeneous" and the surrounding conditional spreading-sequence context.